11 A bus company recorded the ages, in years, of the people on coach A and the people on coach B.

Here are the ages of the 23 people on coach A.

41	42	44	48	52	53	53	53	56	57	57	59
60	61	63	64	64	66	67	69	74	77	79	

(a) Complete the table below to show information about the ages of the people on coach A.

Median	59
Lower quartile	53
Upper quartile	66
Least age	41
Greatest age	79

12th = 59

LQ 6th = 53

UQ 18th = 66

2 marks total
Median 59; LQ 53; UQ 66.

(2)

Here is some information about the ages of the people on coach B.

Median	70
Lower quartile	54
Upper quartile	73
Least age	42
Greatest age	85

Richard says that the people on coach A are younger than the people on coach B.

(b) Is Richard correct?
You must give a reason for your answer.

Yes. Coach A median 59 < coach B median 70.

1 mark
Compare medians to justify younger.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Richard says that the people on coach A vary more in age than the people on coach B.

(c) Is Richard correct?
You must give a reason for your answer.

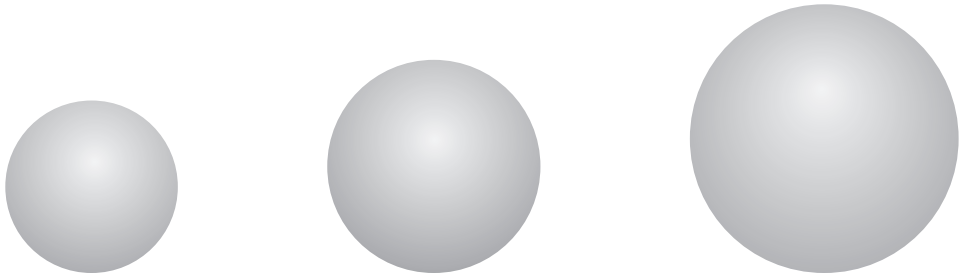
No. Coach A range = 79 – 41 = 38;
coach B range = 85 – 42 = 43.

1 mark
Compare spreads: 38 < 43

(1)

(Total for Question 11 is 4 marks)

12 Here are three spheres.



P **Q** **R**

The volume of sphere **Q** is 50% more than the volume of sphere **P**.
The volume of sphere **R** is 50% more than the volume of sphere **Q**.

Find the volume of sphere **P** as a fraction of the volume of sphere **R**.

(Total for Question 12 is 3 marks)